FC7xxx OSPI User Manual

Rev.0.1

Revision History

Revision	Date	Changes
0.1	2023/12/15	Initial release

Table of Contents

Chapter 1 OSPI Introduction

1.1 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

1.2 Requirement Tracing

OSPI is a Complex Device Driver (CDD), so there are no AUTOSAR requirements regarding this module. It has vendor-specific requirements and implementation.

1.3 Hardware Summary

The OSPI driver is implemented as an Autosar complex device driver. It uses the HyperBus hardware peripherals to provides support for implementing the OSPI protocol.

Hardware and software settings can be configured using an Autosar standard configuration tool. The information required for an OSPI data transfer will be configured in a data structure that will be sent as parameter to the API of the driver.

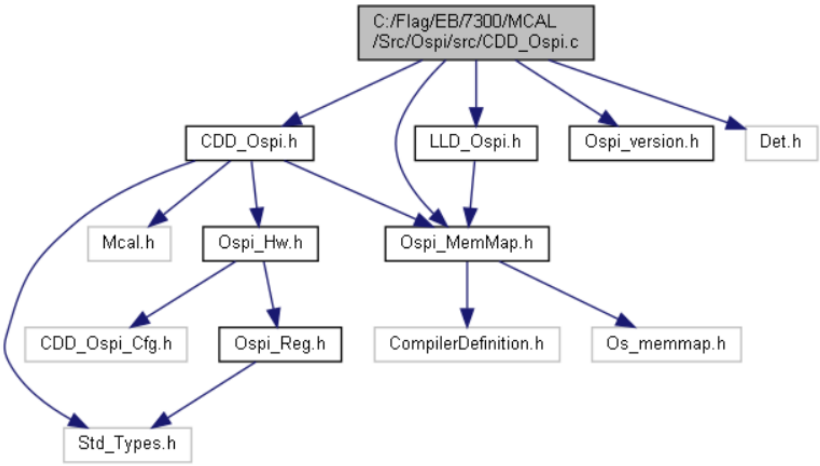
The driver reports errors to the error manager as defined in AUTOSAR.

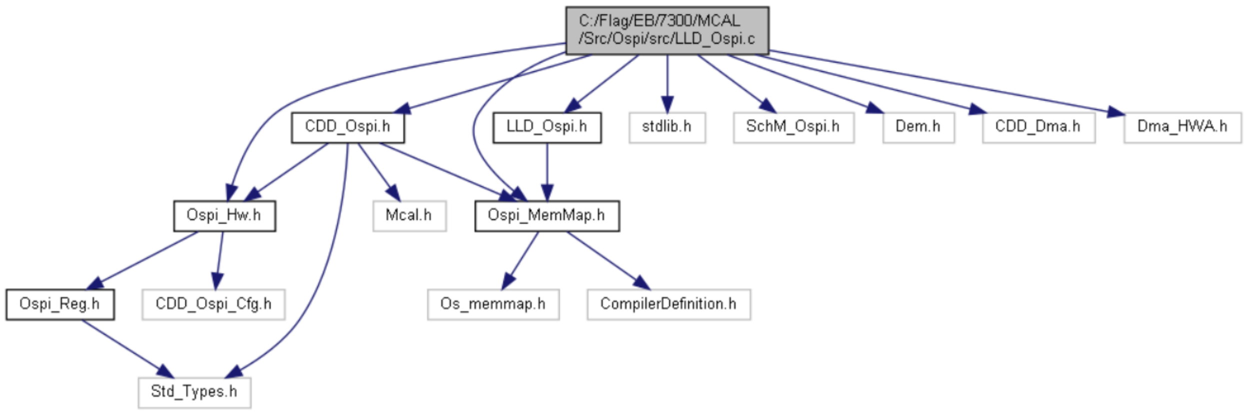
Chapter 2 Software Design

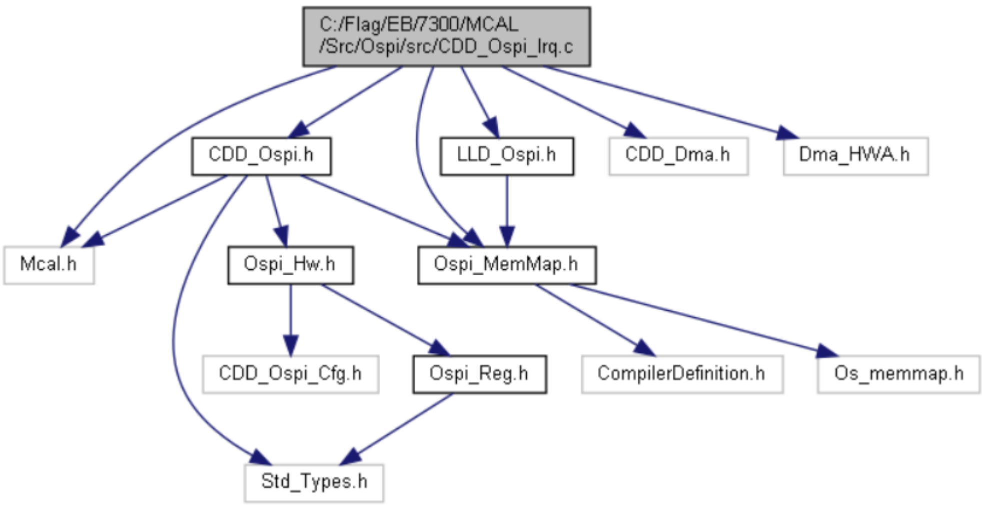
2.1 Rejected Requirements

None

2.2 File Structure







2.3 Macros

2.3.1 Macros in CDD_Ospi.h

- #define **OSPI_INIT_ID** ((uint8) 0x00u)
  Service ID (APIs) for Ospi_Init

- #define **OSPI_GETSTATUS_ID** ((uint8) 0x01u)
  Service ID (APIs) for Ospi_DeInit

- #define **OSPI_DEINIT_ID** ((uint8) 0x02u)
  Service ID (APIs) for Ospi_Init

- #define **OSPI_SET_FLASH_CONFIG_ID** ((uint8) 0x03u)
  Service ID (APIs) for Ospi_SetFlashConfig.

- #define **OSPI_UPDATE_LUT_ID** ((uint8) 0x04u)
  Service ID (APIs) for Ospi_UpdateLUT.

- #define **OSPI_SYNC_READ_ID** ((uint8) 0x05u)
  Service ID (APIs) for Ospi_HyperBusSyncCommandRead.

- #define **OSPI_SYNC_WRITE_ID** ((uint8) 0x06u)
  Service ID (APIs) for Ospi_HyperBusSyncCommandWrite.

- #define **OSPI_SET_FIFO_TxWATERMARK_ID** ((uint8) 0x07u)
  Service ID (APIs) for Ospi_HyperBusSyncCommandSetTxFifoWatermark.

- #define **OSPI_SET_FIFO_RxWATERMARK_ID** ((uint8) 0x08u)
  Service ID (APIs) for Ospi_LLD_HyperBusSyncCommandSetRxFifoWatermark.

- #define **OSPI_CONFIG_DMA_ID** ((uint8)0x09u)
  Service ID (APIs) for Ospi_HyperBusSyncCommandDmaInit.

- #define **OSPI_ENABLE_TX_DMA_ID** ((uint8)0x10u)
  Service ID (APIs) for Ospi_HyperBusSyncCommandeEnableTxDma.

- #define **OSPI_ENABLE_RX_DMA_ID** ((uint8)0x11u)
  Service ID (APIs) for Ospi_HyperBusSyncCommandeEnableRxDma.

- #define **OSPI_SYNC_DMA_READ_ID** ((uint8) 0x11u)
  Service ID (APIs) for Ospi_HyperBusSyncCommandDmaRead.

- #define **OSPI_SYNC_DMA_WRITE_ID** ((uint8) 0x12u)
  Service ID (APIs) for Ospi_HyperBusSyncCommandDmaWrite.

- #define **OSPI_E_ALREADY_INITIALIZED** ((uint8)0x01u)
  Service ID (APIs) for Det reporting.

- #define **OSPI_E_UNINIT** ((uint8)0x02u)
  Service ID (APIs) for Det reporting.

- #define **OSPI_E_PARAM_POINTER** ((uint8)0x03u)
  Service ID (APIs) for Det reporting.

- #define OSPI_E_PARAM_OUTRANGE ((uint8)0x04u)
  Service ID (APIs) for Det reporting.

2.3.2 Macros in Ospi_version.h

- #define **OSPI_AR_RELEASE_MAJOR_VERSION** 4
- #define **OSPI_AR_RELEASE_MINOR_VERSION** 6
- #define **OSPI_AR_RELEASE_REVISION_VERSION** 0
- #define **OSPI_SW_MAJOR_VERSION** 0
- #define **OSP_SW_MINOR_VERSION** 2
- #define **OSPI_SW_PATCH_VERSION** 0
- #define **OSPI_VENDOR_ID** 174
- #define **OSPI_MODULE_ID** 255

2.3.3 Macros in LLD_Ospi.h

- #define **SINGLE_IO** 0x00
- #define **DUAL_IO** 0x01
- #define **QUAD_IO** 0x02
- #define **OCTAL_IO** 0x03
- #define **OSPI_CMD_STOP** 0x0
- #define **OSPI_CFG_DRV** 0x1

- #define **OSPI_CFG_ADDR** 0x2
- #define **OSPI_CMD_DUMMY** 0x3
- #define **OSPI_CFG_MODE8** 0x4
- #define **OSPI_CFG_MODE2** 0x5
- #define **OSPI_CFG_MODE4** 0x6
- #define **OSPI_READ_DRV** 0x7
- #define **OSPI_WRITE_DRV** 0x8
- #define **OSPI_CMD_END** 0x9
- #define **OSPI_CFG_ADDR_DDR** 0xA
- #define **OSPI_CFG_MODE8_DDR** 0xB
- #define **OSPI_CFG_MODE2_DDR** 0xC
- #define **OSPI_CFG_MODE4_DDR** 0xD
- #define **OSPI_READ_DRV_DDR** 0xE
- #define **OSPI_WRITE_DRV_DDR** 0xF
- #define **OSPI_CFG_DRV_DDR** 0x11
- #define **OSPI_CFG_CADDR** 0x12
- #define **OSPI_CFG_CADDR_DDR** 0x13
- #define **OSPI_LUT_SEQ**(cmd0, pad0, op0, cmd1, pad1, op1)

2.3.4 Macros in CDD_Ospi_Cfg.h

- #define **OSPI_AR_RELEASE_MAJOR_VERSION** 4
- #define **OSPI_AR_RELEASE_MINOR_VERSION** 3
- #define **OSPI_AR_RELEASE_REVISION_VERSION** 1

- #define **OSPI_SW_MAJOR_VERSION** 1
- #define **OSP_SW_MINOR_VERSION** 1
- #define **OSPI_SW_PATCH_VERSION** 0
- #define **OSPI_VENDOR_ID** 174
- #define **OSPI_MODULE_ID** 255
- #**define OSPI_DEV_ERROR_DETECT** (STD_ON)
  Switches the Development Error functionality ON or OFF.
- #define **OSPI_DIS_DEM_REPORT_ERR_STAT** (STD_OFF)
  Switches the Production Error functionality ON or OFF.
- #define **OSPI_ALLOW_LEN_LINE_TUNNING_AT_INIT** (STD_OFF)
  Switches the Len Line tuning at initialization.
- #define **OSPI_HYPERBUS_HANDLING_ALLOWED** (STD_ON)
  Switches the Interruptible Sequences handling functionality ON or OFF.
- #define **OSPI_REGULAR_COMMAND_HANDLING_ALLOWED** (STD_OFF)
  Enables code related to Regular command protocol memories.
- #define **OSPI_DMA_HANDLING_ALLOWED** (STD_ON)
  Enables code related to DMA.
- #define **OSPI_MULTICORE_SUPPORT** (STD_OFF)
  Enables the Multicore Support.

2.3.5 Macros in Ospi_Hw.h

- #define **OSPI0_MEMMAP_BASE_ADDR** 0x68000000
- #define **OSPI0_MEMMAP_TOP_ADDR** 0x6A000000
- #define **CTRL_RST_VALUE** 0x34000

2.4 Enums

2.4.1 Enumerations in CDD_Ospi.h

2.4.1.1 Ospi_StatusType

Enumeration	Ospi_StatusType

Description	States that defines the controllers.	
Values	**Value**	**Description**
	OSPI_IDLE = 0,	Ospi channel is idle
	OSPI_HYPERBUS_INIT = 1,	Ospi hyperbus mode is init
	OSPI_UNINIT = 2,	Ospi is uninit
	OSPI_ERROR = 3,	Ospi is error
	OSPI_SUCCESS = 4,	Ospi is success
	OSPI_TIMEOUT = 5,	Ospi is timeout
	OSPI_BUSY = 6,	Ospi is busy
	OSPI_BUSY_TX=0x30,	
	OSPI_BUSY_RX=0x31	

2.4.1.2 Ospi_CommandType

Enumeration	Ospi_CommandType	
Description	The current command is read or write.	
Values	**Value**	**Description**
	OSPI_COMMAND_READ = 0,	The current command is read
	OSPI_COMMAND_WRITE = 1	The current command is write

2.4.2 Enumerations in Ospi_Hw.h

2.4.2.1 OSPI_DqsSrcSelType

Enumeration	OSPI_DqsSrcSelType	
Description	Type for dqs source	
Values	**Value**	**Description**
	DQS_INTER_LOOPBACK = 0,	None DQS self-loopback loop
	DQS_PAD_LOOPBACK =1,	DQS self-return loops
	DQS_EXTERNAL_PADINPUT =2	DQS comes from an external device

2.4.2.2 OSPI_ClockDivideType

Enumeration	OSPI_ClockDivideType	
Description	Specifies the clock division for OSPI	
Values	**Value**	**Description**
	OSPI_CLOCK_DIV_1 = 0U,	Clock frequency divisions is 1
	OSPI_CLOCK_DIV_2 = 1U,	Clock frequency divisions is 2
	OSPI_CLOCK_DIV_3 = 2U,	Clock frequency divisions is 3
	OSPI_CLOCK_DIV_4 = 3U,	Clock frequency divisions is 4
	OSPI_CLOCK_DIV_5 = 4U,	Clock frequency divisions is 5
	OSPI_CLOCK_DIV_6 = 5U,	Clock frequency divisions is 6
	OSPI_CLOCK_DIV_7 = 6U,	Clock frequency divisions is 7
	OSPI_CLOCK_DIV_8 = 7U	Clock frequency divisions is 8

2.4.2.3 OSPI_ClockMuxType

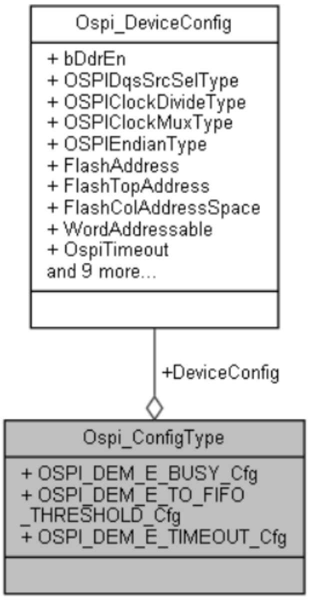
Enumeration	OSPI_ClockMuxType	
Description	Internal Reference Clock Source	
Values	**Value**	**Description**
	OSPI_MUX_PLL0	Internal Reference Clock Source is PLL0
	OSPI_MUX_FIRC	Internal Reference Clock Source is FIRC
	OSPI_MUX_PLL1	Internal Reference Clock Source is PLL1
	OSPI_MUX_BUSCLOCK	Internal Reference Clock Source is bus clock

2.4.2.4 OSPI_EndianType

Enumeration	OSPI_EndianType	
Description	Select the byte order.	
Values	**Value**	**Description**
	OSPI_BIG_ENDIAN = 0,	Big endian
	OSPI_LITTLE_ENDIAN =1	Little endian

2.5 Structures

2.5.1 Ospi_ConfigType

Structure	Ospi_ConfigType
Description	The top level structure containing all the needed parameters for the Ospi Handler/Driver.
Diagram	
Data Fields	• Const Ospi_DeviceConfig Array of LLD OSPI device instances • Mcal_DemErrorType OSPI Driver DEM Error:

2.5.2 Ospi_DeviceConfig

Structure	Ospi_DeviceConfig

Description	The structure holds the Device configuration parameters.
Diagram	
Data Fields	• boolean bDdrEn; • OSPI_DqsSrcSelType OSPIDqsSrcSelType • OSPI_ClockDivideType OSPIClockDivideType • OSPI_ClockMuxType OSPIClockMuxType • OSPI_EndianType OSPIEndianType • uint32 FlashAddress; • uint32 FlashTopAddress; • uint8 FlashColAddressSpace; • uint8 WordAddressable; • uint32 OspiTimeout; • uint8 CsHoldTime; • uint8 CsSetupTime; • uint8 DelayLine; • uint8 u8OspiTxDmaChannel; DMA channel used for transmit data • uint8 u8OspiRxDmaChannel; DMA channel used for receive data* • uint8 OspiFifoThreshold; • Ospi_NotifyType *Ospi_TransferCompleteNotification; • Ospi_NotifyType *Ospi_TransferErrorNotification;

2.5.3 OSPI_Type

Structure	OSPI_Type
Description	This structure contains all the needed data to configure one SPI Sequence
Diagram	N/A
Data Fields	• __IO uint32 CTRL Control Register, offset: 0x0 • uint8 RESERVED_0[4]; • __IO uint32 CMDC Command Control Register, offset: 0x8 • __IO uint32 FLS_CFG Flash Configuration Register, offset: 0xC

	• uint8 RESERVED_1[20]; • __IO uint32 SOC_CFG SOC Configuration Register, offset: 0x24 • uint8 RESERVED_2[216]; • __IO uint32 FLS_AR Flash Address Register, offset: 0x100 • __IO uint32 FLS_CAR Flash Column Address Register, offset: 0x104 • __IO uint32 SAMP_CTRL Sampling Control Register, offset: 0x108 • __IO uint32 RFSR Receive FIFO Status Register, offset: 0x10C • __IO uint32 RF_CFG Receive FIFO Configuration Register, offset: 0x110 • uint8 RESERVED_3[12]; • __IO uint32 RFD2R Receive FIFO Data Secondary Register, offset: 0x120 • uint8 RESERVED_4[44]; • __IO uint32 TFSR Transmit FIFO Status Register, offset: 0x150 • __IO uint32 TFDR Transmit FIFO Data Register, offset: 0x154 • __IO uint32 TF_CFG Transmit FIFO Configuration Register, offset: 0x158 • __IO uint32 STATUS Status Register, offset: 0x15 • __IO uint32 FLAG Flag Register, offset: 0x160

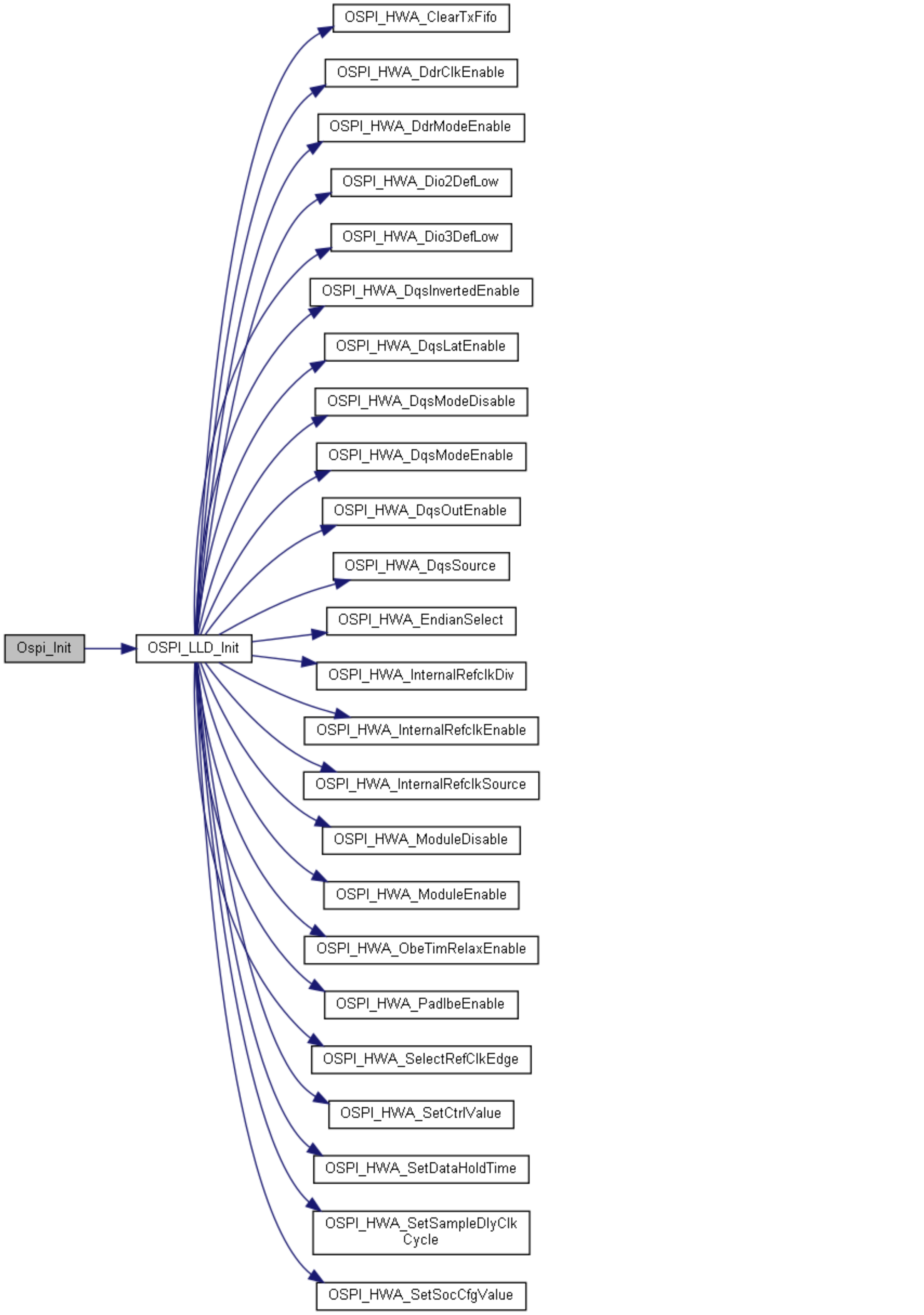
	• __IO uint32 IND_EN Interrupt and DMA Enable Register, offset: 0x164 • uint8 RESERVED_5[24]; • __IO uint32 FLS_TAR Flash Top Address Register, offset: 0x180 • uint8 RESERVED_6[124]; • __IO uint32 RFDR[OSPI_RFDR_COUNT] Receive FIFO Data Register, offset: 0x200 • uint8 RESERVED_7[128]; • __IO uint32 LUT_KEY Lookup Table Key Register, offset: 0x300 • __IO uint32 LUT_CFG Lookup Table Configuration Register, offset: 0x304 • uint8 RESERVED_8[8]; • __IO uint32 LUT[OSPI_LUT_COUNT] Lookup Table Register, offset: 0x310

2.6 API Functions

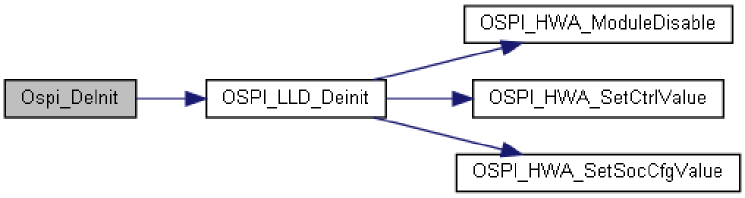
2.6.1 Functions in CDD_Ospi.c

2.6.1.1 void Ospi_Init (const Ospi_ConfigType *ConfigPtr)

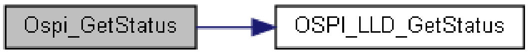
Function	Ospi_Init (const Ospi_ConfigType *ConfigPtr)
Description	This function initializes the OSPI hardware unit and the driver.

Diagram		
Parameters	**Parameter**	**Description**
	ConfigPtr	Specifies the pointer to the configuration set
Returns	N/A	

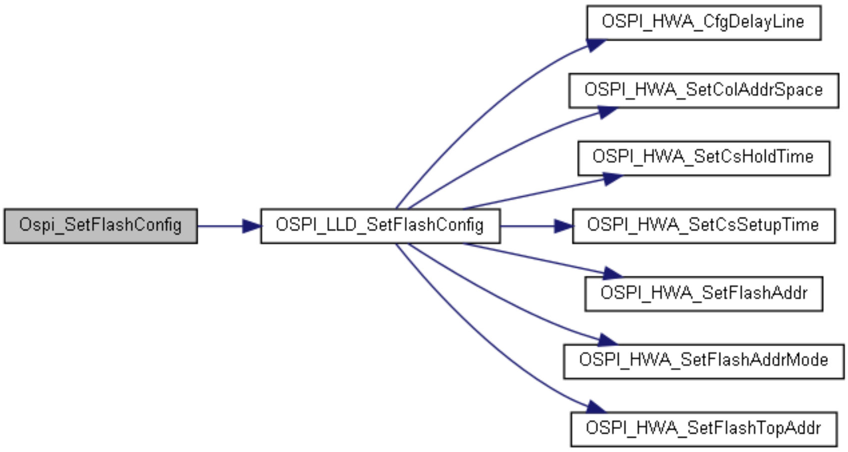
2.6.1.2 void Ospi_DeInit(void)

Function	Ospi_DeInit(void)	
Description	This function de-initializes the OSPI driver.	
Diagram		
Parameters	**Parameter**	**Description**
	N/A	N/A
Returns	N/A	

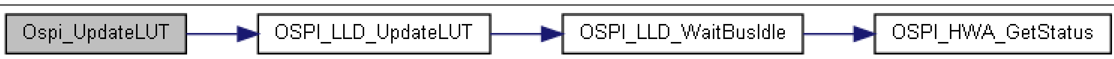
2.6.1.3 Ospi_StatusType Ospi_GetStatus(void)

Function	Ospi_GetStatus(void)	
Description	This function returns the status of the OSPI driver.	
Diagram		
Parameters	**Parameter**	**Description**
	N/A	N/A
Returns	Ospi_StatusType	

2.6.1.4 Std_ReturnType Ospi_SetFlashConfig (void)

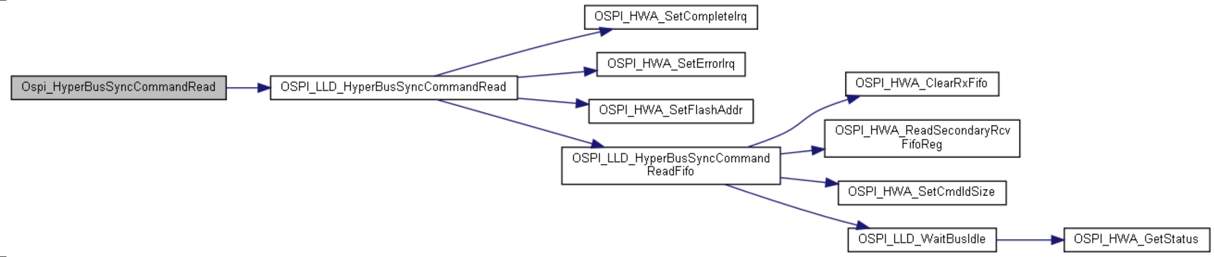
Function	Ospi_SetFlashConfig (void)	
Description	This function sets the flash parameter.	
Diagram		
Parameters	**Parameter**	**Description**
	N/A	N/A
Returns	Std_ReturnType	

2.6.1.5 Std_ReturnType Ospi_UpdateLUT (uint32 index, const uint32 * cmd, uint32 count)

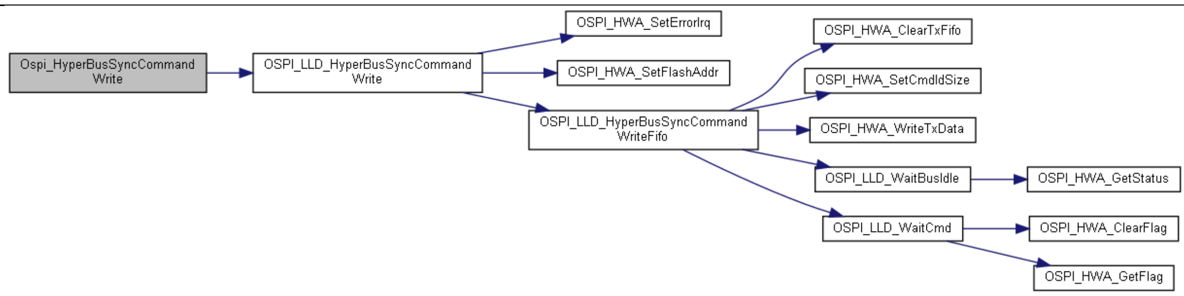
Function	Ospi_UpdateLUT(uint32 index, const uint32 * cmd, uint32 count)	
Description	This function update the LUT.	
Diagram		
Parameters	**Parameter**	**Description**

	index	Index to be written
	cmd	Command sequence array
	count	Number of sequences
Returns	Std_ReturnType	

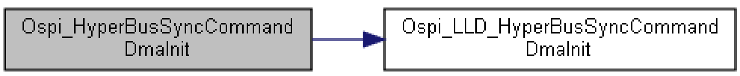
2.6.1.6 Ospi_StatusType Ospi_HyperBusSyncCommandRead(uint8 u8SeqId, uint32 *pBuf, uint32 u32Size)

Function	Ospi_HyperBusSyncCommandRead(uint8 u8SeqId, uint32 *pBuf, uint32 u32Size)	
Description	This function is used for synchronous command with read data.	
Diagram		
Parameters	**Parameter**	**Description**
	u8SeqId	the cmd id location in lut
	pBuf	the read buffer start address
	u32Size	data size to be read
Returns	Ospi_StatusType	

2.6.1.7 Ospi_StatusType Ospi_HyperBusSyncCommandWrite(uint8 u8SeqId, uint32 *pBuf, uint32 u32Size)

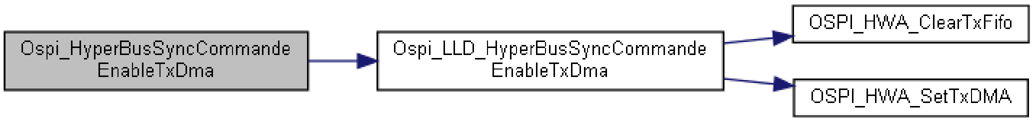
Function	Ospi_HyperBusSyncCommandWrite(uint8 u8SeqId, uint32 *pBuf, uint32 u32Size)	
Description	This function is used for synchronous command with write data.	
Diagram		
Parameters	**Parameter**	**Description**
	u8SeqId	the cmd id location in lut.
	pBuf	the write buffer start address.
	u32Size	data size to be written.
Returns	Ospi_StatusType	

2.6.1.8 Std_ReturnType Ospi_HyperBusSyncCommandDmaInit(const uint32 * writeBuf, uint32 *readBuf, uint16 testsize)

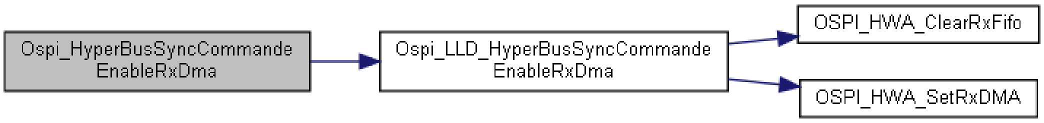
Function	Ospi_HyperBusSyncCommandDmaInit (const uint32 * writeBuf, uint32 *readBuf, uint16 testsize)	
Description	This function initializes the OSPI driver in DMA mode.	
Diagram		
Parameters	**Parameter**	**Description**
	writeBuf	Pointer to write array

	readBuf	Pointer to read array
	testsize	Transmission size
Returns	Std_ReturnType	

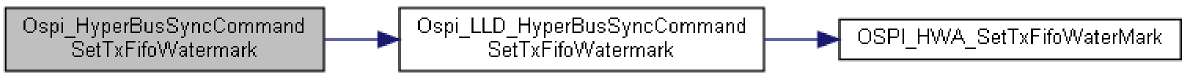
2.6.1.9 Std_ReturnType Ospi_HyperBusSyncCommandeEnableTxDma (void)

Function	Ospi_HyperBusSyncCommandeEnableTxDma (void)	
Description	This function enables dma tx mode.	
Diagram		
Parameters	**Parameter**	**Description**
	N/A	N/A
Returns	Std_ReturnType	

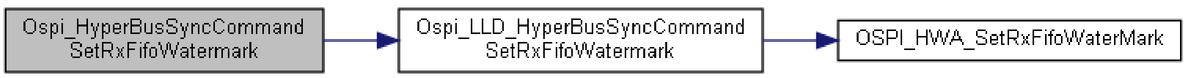
2.6.1.10 Std_ReturnType Ospi_HyperBusSyncCommandeEnableRxDma (void)

Function	Ospi_HyperBusSyncCommandeEnableRxDma (void)	
Description	This function enables dma rx mode.	
Diagram		
Parameters	**Parameter**	**Description**
	N/A	N/A
Returns	Std_ReturnType	

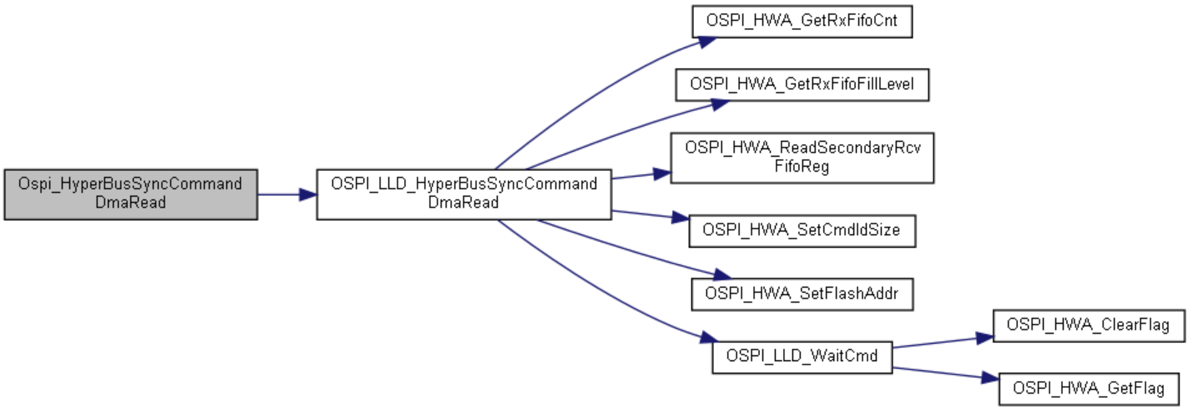
2.6.1.11 Std_ReturnType Ospi_HyperBusSyncCommandSetTxFifoWatermark (void)

Function	Ospi_HyperBusSyncCommandSetTxFifoWatermark (void)	
Description	This function sets the watermark of the transmitted fifo.	
Diagram		
Parameters	**Parameter**	**Description**
	N/A	N/A
Returns	Std_ReturnType	

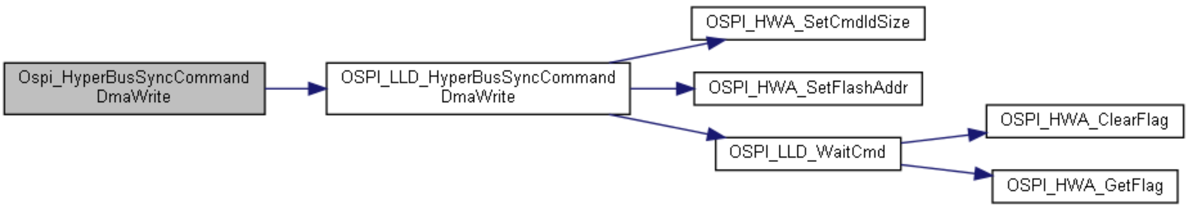
2.6.1.12 Std_ReturnType Ospi_HyperBusSyncCommandSetRxFifoWatermark (void)

Function	Ospi_HyperBusSyncCommandSetRxFifoWatermark (void)	
Description	This function sets the watermark of the reveiving fifo.	
Diagram		
Parameters	**Parameter**	**Description**
	N/A	N/A
Returns	Std_ReturnType	

2.6.1.13 Ospi_StatusType Ospi_HyperBusSyncCommandDmaRead (uint8 u8SeqId, uint32 dataSize, uint32 *readBuf)

Function	Ospi_HyperBusSyncCommandDmaRead (uint8 u8SeqId, uint32 dataSize, uint32 *readBuf)	
Description	This function is used for synchronous command with read data in dma mode.	
Diagram		
Parameters	**Parameter**	**Description**
	u8SeqId	the cmd id location in lut.
	readBuf	Pointer to read array.
	dataSize	data size to be read.
Returns	Ospi_StatusType	

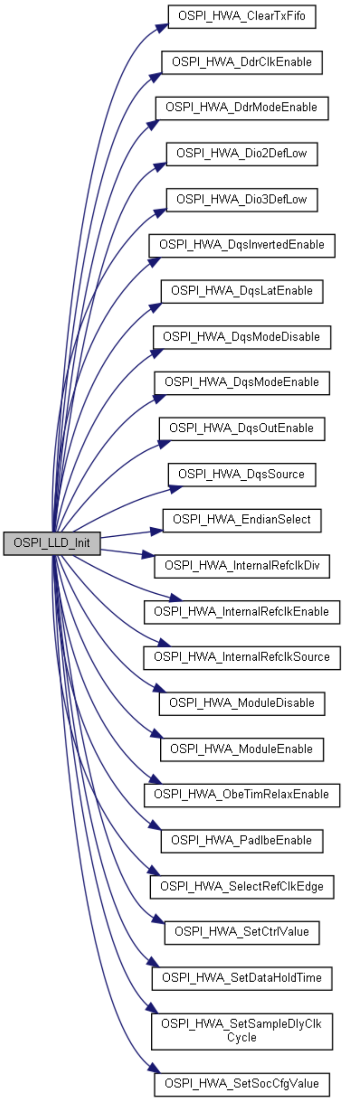
2.6.1.14 Ospi_StatusType Ospi_HyperBusSyncCommandDmaWrite (uint8 u8SeqId, uint32 dataSize)

Function	Ospi_HyperBusSyncCommandDmaWrite (uint8 u8SeqId, uint32 dataSize)	
Description	This function is used for synchronous command with write data in dma mode.	
Diagram		
Parameters	**Parameter**	**Description**
	u8SeqId	the cmd id location in lut.
	dataSize	data size to be written.
Returns	Ospi_StatusType	

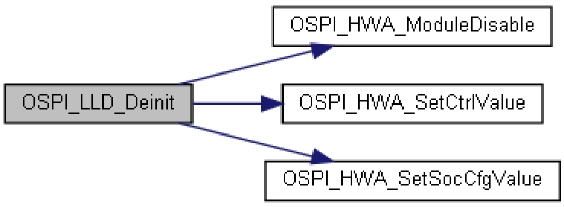
2.6.2 Functions in LLD_Ospi.c

2.6.2.1 void OSPI_LLD_Init (void)

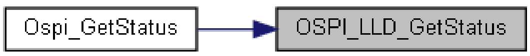
Function	Ospi_LLD_Init (void)
Description	Low-level initialize function. Initialize OSPI configuration.

Diagram	 	
Parameters	**Parameter**	**Description**
	N/A	N/A

2.6.2.2 void Ospi_LLD_DeInit(void)

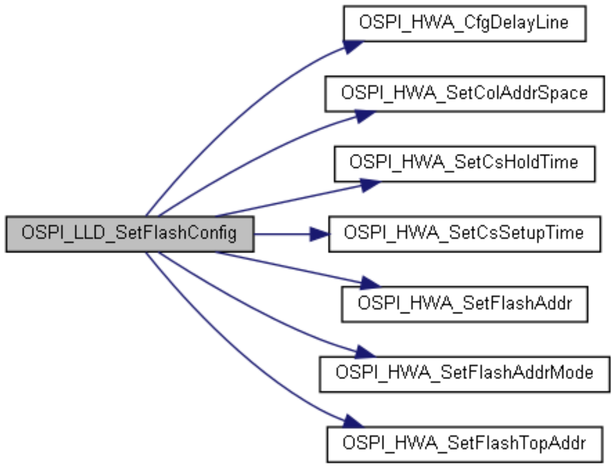
Function	Ospi_LLD_DeInit(void)	
Description	DeInitialize and set to reset valmue all registers.	
Diagram	 	
Parameters	**Parameter**	**Description**
	N/A	N/A

2.6.2.3 Ospi_StatusType Ospi_LLD_GetStatus(void)

Function	Ospi_LLD_GetStatus(void)
Description	Low-level get status of OSPI.
Diagram	

Parameters	**Parameter**	**Description**
	N/A	N/A
Returns	Ospi_StatusType	

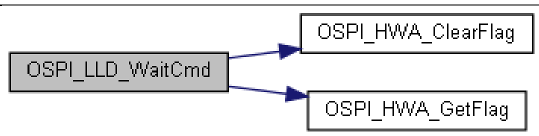
2.6.2.4 Std_ReturnType Ospi_LLD_SetFlashConfig (void)

Function	Ospi_LLD_SetFlashConfig (void)	
Description	Low-level OSPI config flash parameter. OSPI Config Flash Parameter. Confiugration for the ospi peripherals device options.	
Diagram		
Parameters	**Parameter**	**Description**
	N/A	N/A
Returns	Std_ReturnType	

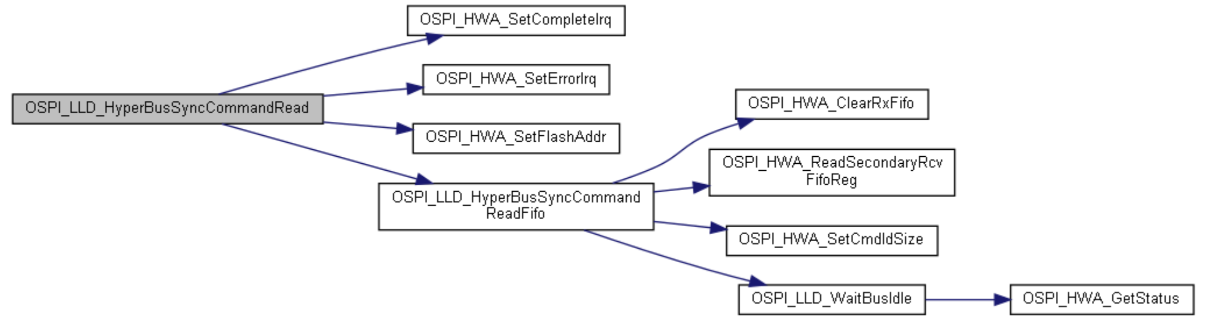
2.6.2.5 Std_ReturnType Ospi_LLD_UpdateLUT (uint32 index, const uint32 * cmd, uint32 count)

Function	Ospi_LLD_UpdateLUT (uint32 index, const uint32 * cmd, uint32 count)	
Description	Low-level OSPI config LUT.	
Diagram		
Parameters	**Parameter**	**Description**
	index	Index to be written
	cmd	Command sequence array
	count	Number of sequences
Returns	Std_ReturnType	

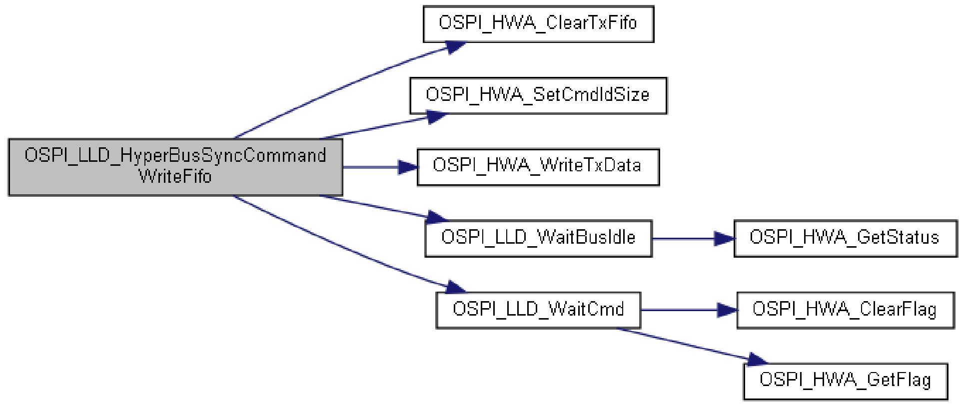
2.6.2.6 Std_ReturnType Ospi_LLD_WaitCmd (void)

Function	Ospi_LLD_WaitCmd (void)	
Description	Low-level OSPI config LUT.	
Diagram		
Parameters	**Parameter**	**Description**
	N/A	N/A
Returns	Std_ReturnType	

2.6.2.7 Ospi_StatusType Ospi_LLD_HyperBusSyncCommandRead(uint8 u8SeqId, uint32 *pBuf, uint32 u32Size)

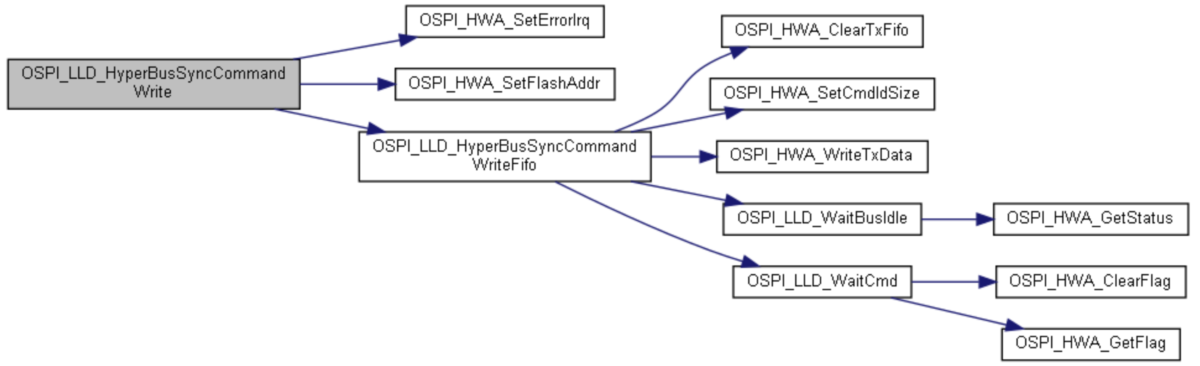
Function	Ospi_LLD_HyperBusSyncCommandRead(uint8 u8SeqId, uint32 *pBuf, uint32 u32Size)	
Description	OSPI read sequence data.	
Diagram		
Parameters	**Parameter**	**Description**
	u8SeqId	the cmd id location in lut.
	pBuf	the read buffer start address.
	u32Size	data size to be read
Returns	Ospi_StatusType	

2.6.2.8 Ospi_StatusType Ospi_LLD_HyperBusSyncCommandReadFifo(uint8 u8SeqId, uint32 *pBuf, uint8 u8Size)

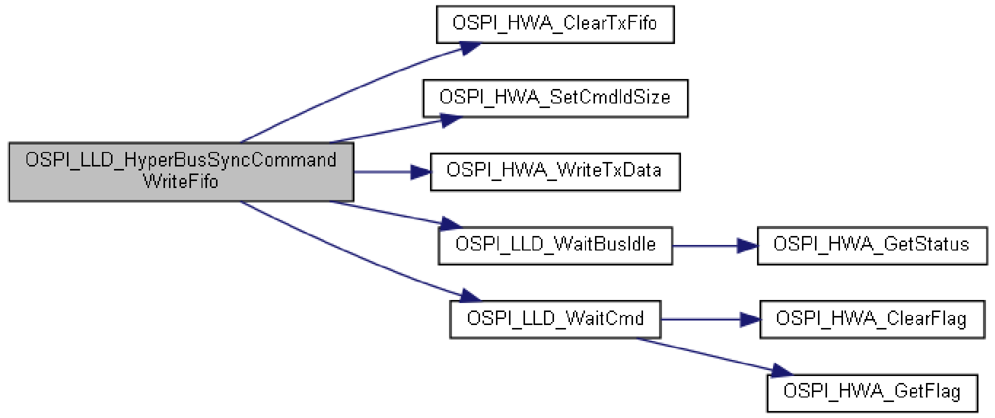
Function	Ospi_LLD_HyperBusSyncCommandRead(uint8 u8SeqId, uint32 *pBuf, uint8 u8Size)	
Description	OSPI read fifo. Fifo size is 16words.	
Diagram		
Parameters	**Parameter**	**Description**
	u8SeqId	the cmd id location in lut.
	pBuf	the read buffer start address.
	u8Size	fifo size to be read
Returns	Ospi_StatusType	

2.6.2.9 Ospi_StatusType Ospi_LLD_HyperBusSyncCommandWrite(uint8 u8SeqId, uint32 *pBuf, uint32 u32Size)

Function	Ospi_LLD_HyperBusSyncCommandWrite(uint8 u8SeqId, uint32 *pBuf, uint32 u32Size)
Description	OSPI write sequence data.

Diagram		
Parameters	**Parameter**	**Description**
	u8SeqId	the cmd id location in lut.
	pBuf	the write buffer start address.
	u32Size	data size to be written.
Returns	Ospi_StatusType	

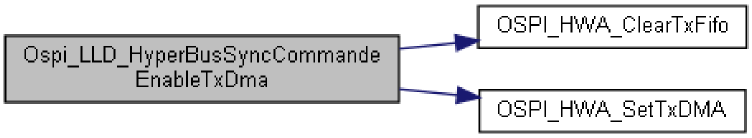
2.6.2.10 Ospi_StatusType Ospi_LLD_HyperBusSyncCommandWriteFifo(uint8 u8SeqId, uint32 *pBuf, uint8 u8Size)

Function	Ospi_LLD_HyperBusSyncCommandWriteFifo (uint8 u8SeqId, uint32 *pBuf, uint8 u8Size)	
Description	OSPI write fifo. Fifo size is 16words.	
Diagram		
Parameters	**Parameter**	**Description**
	u8SeqId	the cmd id location in lut.
	pBuf	the write buffer start address.
	u8Size	fifo size to be written.
Returns	Ospi_StatusType	

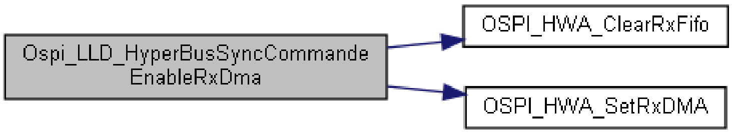
2.6.2.11 void Ospi_LLD_HyperBusSyncCommandDmaInit(const uint32 * writeBuf, uint32 *readBuf, uint16 testsize)

Function	Ospi_LLD_HyperBusSyncCommandDmaInit (const uint32 * writeBuf, uint32 *readBuf, uint16 testsize)	
Description	Initialize DMA Tx/Rx cahnnels for OSPI Hw unit.	
Diagram	N/A	
Parameters	**Parameter**	**Description**
	writeBuf	Pointer to write array
	readBuf	Pointer to read array
	testsize	Transmission size
Returns	N/A	

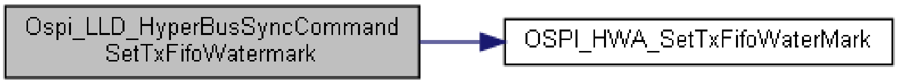
2.6.2.12 void Ospi_LLD_HyperBusSyncCommandeEnableTxDma (void)

Function	Ospi_LLD_HyperBusSyncCommandeEnableTxDma (void)	
Description	low-level enables dma tx mode	
Diagram		
Parameters	**Parameter**	**Description**
	N/A	N/A
Returns	N/A	

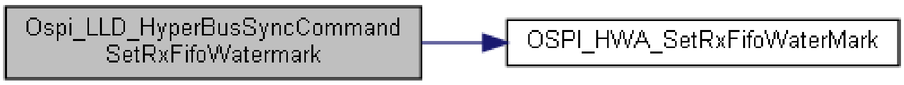
2.6.2.13 void Ospi_LLD_HyperBusSyncCommandeEnableRxDma (void)

Function	Ospi_LLD_HyperBusSyncCommandeEnableRxDma (void)	
Description	This function enables dma rx mode.	
Diagram		
Parameters	**Parameter**	**Description**
	N/A	N/A
Returns	N/A	

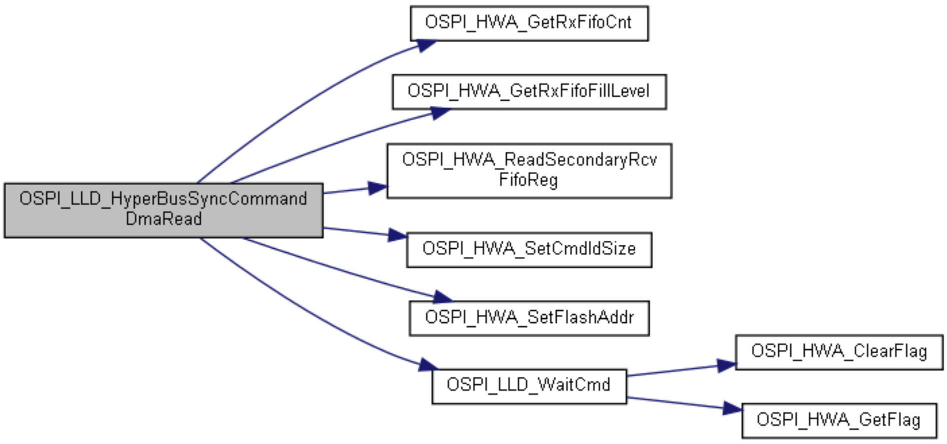
2.6.2.14 Std_ReturnType Ospi_LLD_HyperBusSyncCommandSetTxFifoWatermark (void)

Function	Ospi_LLD_HyperBusSyncCommandSetTxFifoWatermark (void)	
Description	Set transmit fifo watermark. Indicates how many valid entries will trigger a transmit action.	
Diagram		
Parameters	**Parameter**	**Description**
	N/A	N/A
Returns	Std_ReturnType	

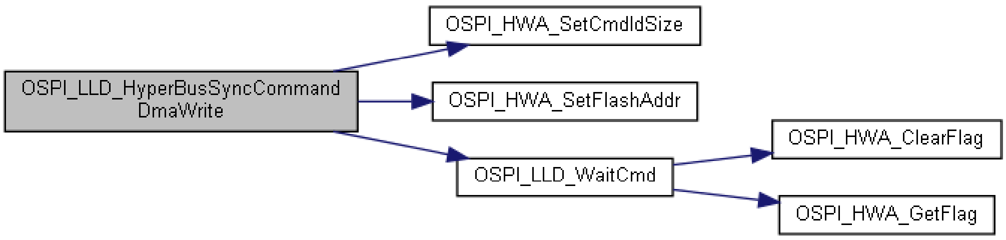
2.6.2.15 Std_ReturnType Ospi_LLD_HyperBusSyncCommandSetRxFifoWatermark (void)

Function	Ospi_LLD_HyperBusSyncCommandSetRxFifoWatermark (void)	
Description	Set receive fifo watermark. Indicates how many valid entries will trigger a readout action.	
Diagram		
Parameters	**Parameter**	**Description**
	N/A	N/A
Returns	Std_ReturnType	

2.6.2.16 Ospi_StatusType Ospi_LLD_HyperBusSyncCommandDmaRead (uint8 u8SeqId, uint32 dataSize, uint32 *readBuf)

Function	Ospi_LLD_HyperBusSyncCommandDmaRead (uint8 u8SeqId, uint32 dataSize, uint32 *readBuf)	
Description	The function that triggers dma read.	
Diagram		
Parameters	**Parameter**	**Description**
	u8SeqId	the cmd id location in lut.
	readBuf	Pointer to read array.
	dataSize	data size to be read.
Returns	Ospi_StatusType	

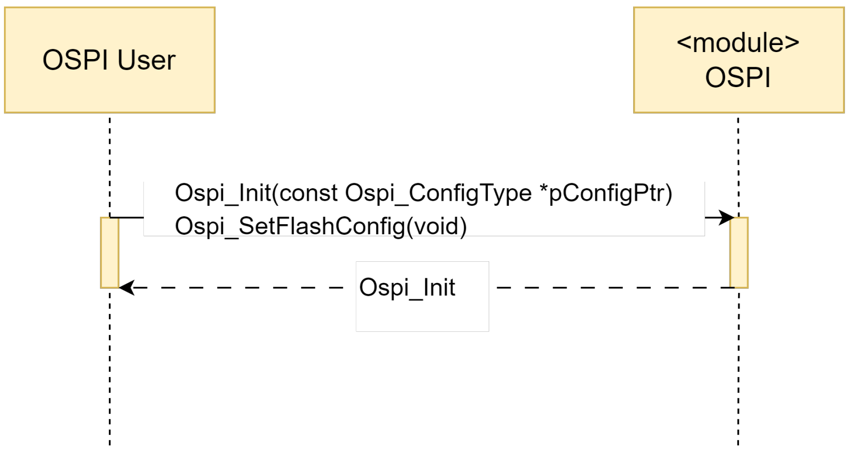
2.6.2.17 Ospi_StatusType Ospi_LLD_HyperBusSyncCommandDmaWrite (uint8 u8SeqId, uint32 dataSize)

Function	Ospi_LLD_HyperBusSyncCommandDmaWrite (uint8 u8SeqId, uint32 dataSize)	
Description	The function that triggers dma write.	
Diagram		
Parameters	**Parameter**	**Description**
	u8SeqId	the cmd id location in lut.
	dataSize	data size to be written.
Returns	Ospi_StatusType	

2.7 API Sequence Diagram

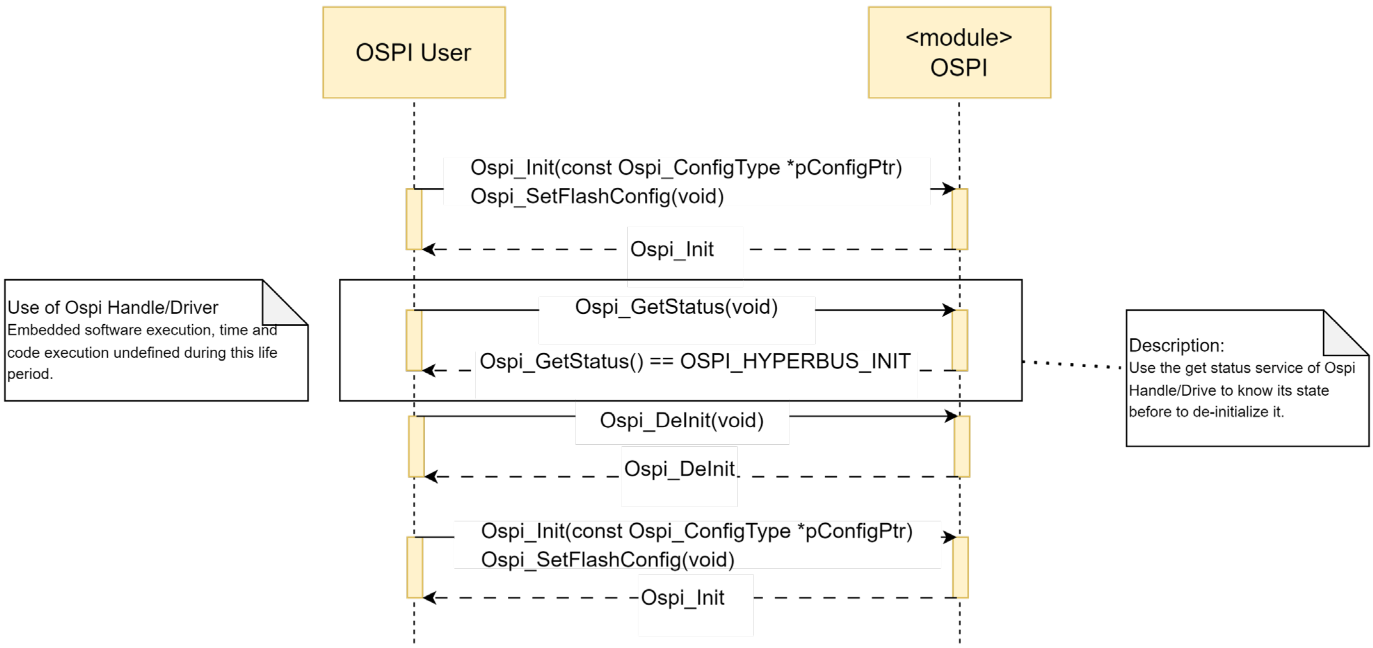
2.7.1 Initialization

The ECU State Manager (EcuM) is responsible for calling the initialization function.

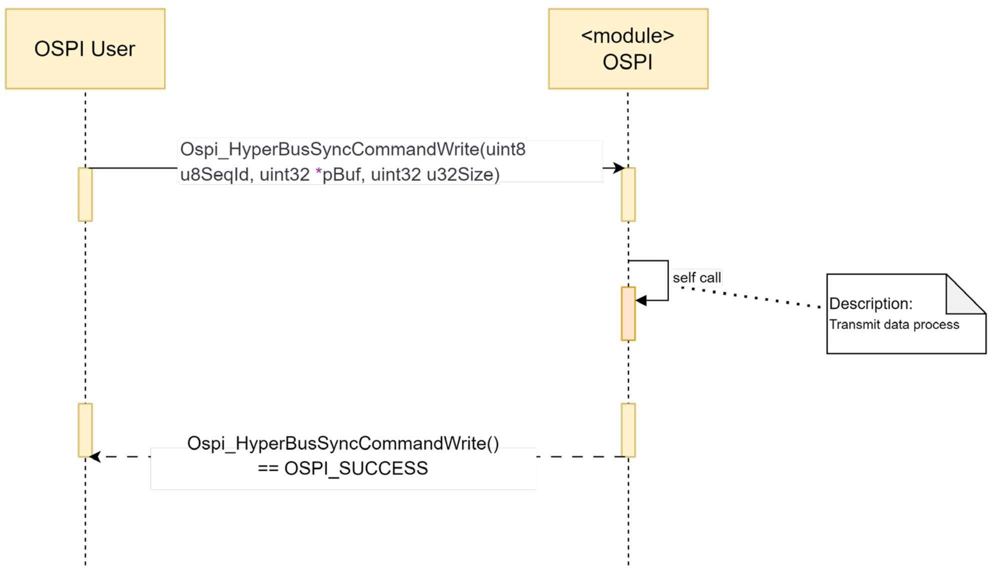


2.7.2 Modes Transitions

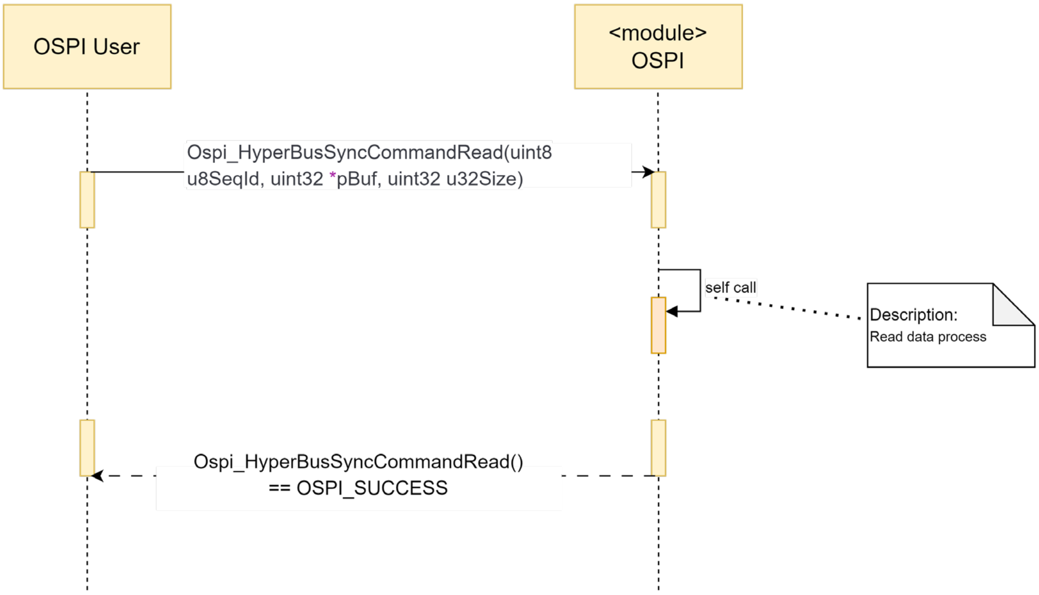
The following sequence diagram shows an example of an Init / DeInit calls for a running mode transition.



2.7.3 SyncWrite



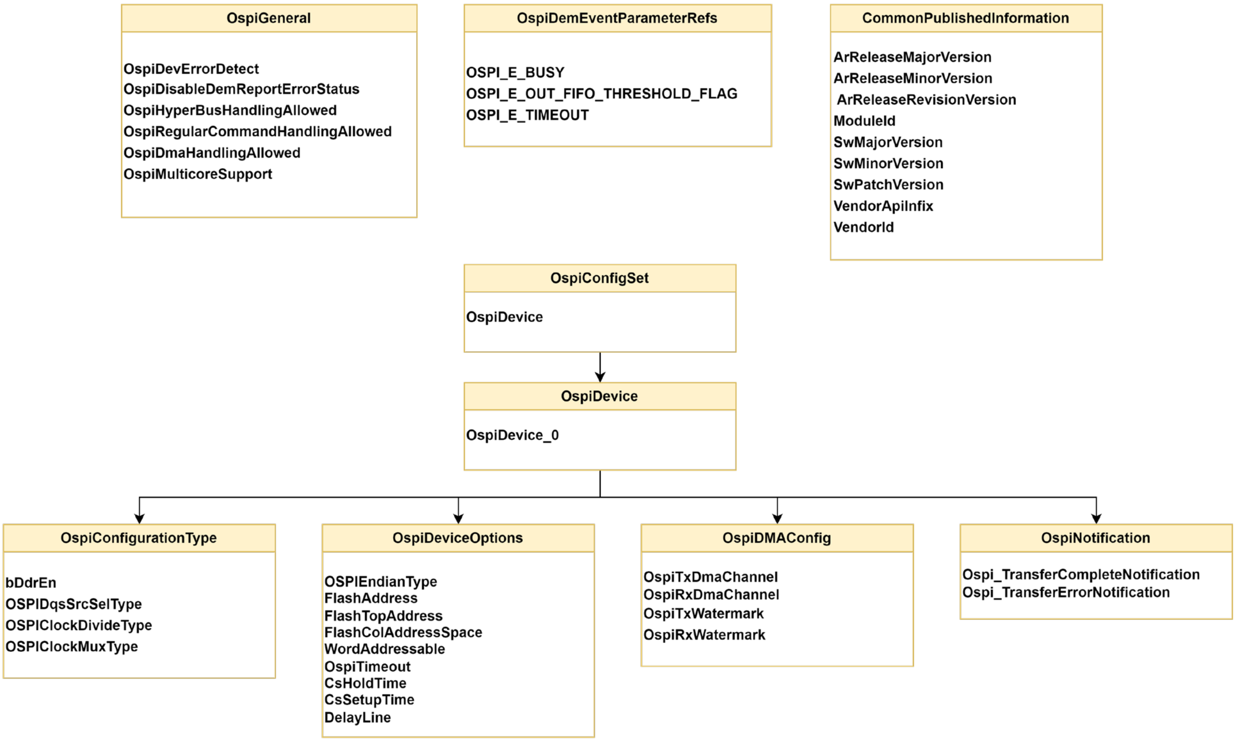
2.7.4 SyncReceive



Chapter 3 Tresos Configuration Items

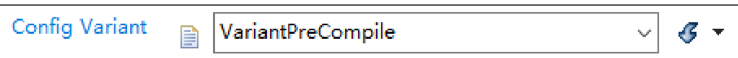
3.1 Container Inclusion Relation

The contain inclusion relation is shown as below:



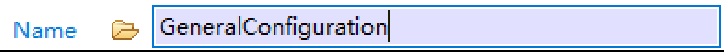
3.2 Containers and Variables

3.2.1 IMPLEMENTATION_CONFIG_VARIANT

Container	IMPLEMENTATION_CONFIG_VARIANT	
Description	N/A	
Screenshot		
Properties	**Property**	**Value**
	Type	ENUMERATION
	Label	Config Variant
	Default	VariantPostBuild
	Range	VariantPostBuild VariantPreCompile
Returns	N/A	

3.2.2 GeneralConfiguration

Container	GeneralConfiguration
Description	General configuration (parameters) of the Ospi Driver software module.

Screenshot		
Properties	**Property**	**Value**
	Type	Container

3.2.2.1 OspiDevErrorDetect

Container	OspiDevErrorDetect	
Description	Switches the Development Error Detection and Notification ON or OFF.	
Screenshot		
Properties	**Property**	**Value**
	Type	BOOLEAN
	Origin	Flagchip
	Default	TRUE

3.2.2.2 OspiDisableDemReportErrorStatus

Container	OspiDisableDemReportErrorStatus	
Description	Switches the production error detection and notification on or off.	
Screenshot		
Properties	**Property**	**Value**
	Type	BOOLEAN
	Origin	Flagchip
	Default	TRUE

3.2.2.3 OspiHyperBusHandlingAllowed

Container	OspiHyperBusHandlingAllowed	
Description	Enables code related to HyperBus protocole management.	
Screenshot		
Properties	**Property**	**Value**
	Type	BOOLEAN
	Origin	Flagchip
	Default	TRUE

3.2.2.4 OspiRegularCommandHandlingAllowed

Container	OspiRegularCommandHandlingAllowed	
Description	Enables code related to Regular command protocole management.	
Screenshot		
Properties	**Property**	**Value**
	Type	BOOLEAN
	Origin	Flagchip
	Default	FALSE

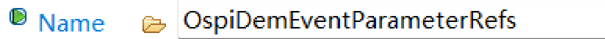
3.2.2.5 OspiDmaHandlingAllowed

Container	OspiDmaHandlingAllowed	
Description	Enables code related to DMA management.	
Screenshot		
Properties	**Property**	**Value**
	Type	BOOLEAN
	Origin	Flagchip
	Default	FALSE

3.2.2.6 OspiMulticoreSupport

Container	OspiMulticoreSupport	
Description	When this parameter is enabled, multi-core feature will be used in OSPI driver.That means mapping the OSPI driver to multiple ECUC partitions to make the module API available in this partition.	
Screenshot		
Properties	**Property**	**Value**
	Type	BOOLEAN
	Origin	Flagchip
	Default	FALSE

3.2.3 OspiDemEventParameterRefs

Container	OspiDemEventParameterRefs	
Description	Container for the references to DemEventParameter elements which shall be invoked using the API Dem_ReportErrorStatus API in case the corresponding error occurs.The EventId is taken from the referenced DemEventParameter's DemEventId value. The standardized errors are provided in the container and can be extended by vendor specific error references.	
Screenshot		
Properties	**Property**	**Value**
	Type	Container

3.2.3.1 OSPI_E_BUSY

Container	OSPI_E_BUSY	
Description	Reference to configured DEM event to report "Ospi Busy".	
Screenshot		
Properties	**Property**	**Value**
	Type	SYMBOLIC-NAME-REFERENCE
	Origin	Flagchip
	SymbolicNameValue	false

3.2.3.2 OSPI_E_OUT_FIFO_THRESHOLD_FLAG

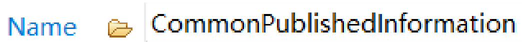
Container	OSPI_E_OUT_FIFO_THRESHOLD_FLAG
Description	Reference to configured DEM event to report "Ospi out of fifo threshold".
Screenshot	

Properties	**Property**	**Value**
	Type	SYMBOLIC-NAME-REFERENCE
	Origin	Flagchip
	SymbolicNameValue	false

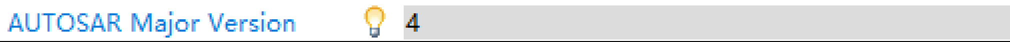
3.2.3.3 OSPI_E_TIMEOUT

Container	OSPI_E_TIMEOUT	
Description	Reference to configured DEM event to report "Ospi timeout".	
Screenshot		
Properties	**Property**	**Value**
	Type	SYMBOLIC-NAME-REFERENCE
	Origin	Flagchip
	SymbolicNameValue	false

3.2.4 CommonPublishedInformation

Container	CommonPublishedInformation	
Description	Common container, aggregated by all modules. It contains published information about vendor and versions.	
Screenshot		
Properties	**Property**	**Value**
	Type	Container

3.2.4.1 ArReleaseMajorVersion

Variable	ArReleaseMajorVersion	
Description	Major version number of AUTOSAR specification on which the appropriate implementation is based.	
Screenshot		
Properties	**Property**	**Value**
	Type	INTEGER
	Origin	Flagchip

3.2.4.2 ArReleaseMinorVersion

Variable	ArReleaseMinorVersion	
Description	Minor version number of AUTOSAR specification on which the appropriate implementation is based.	
Screenshot		
Properties	**Property**	**Value**
	Type	INTEGER
	Origin	Flagchip

3.2.4.3 ArReleaseRevisionVersion

Variable	ArReleaseRevisionVersion
Description	Revision version number of AUTOSAR specification on which the appropriate implementation is based.

Screenshot		
Properties	**Property**	**Value**
	Type	INTEGER
	Origin	Flagchip

3.2.4.4 SwMajorVersion

Variable	SwMajorVersion	
Description	Major version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	**Property**	**Value**
	Type	INTEGER
	Origin	Flagchip

3.2.4.5 SwMinorVersion

Variable	SwMinorVersion	
Description	Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	**Property**	**Value**
	Type	INTEGER
	Origin	Flagchip

3.2.4.6 SwPatchVersion

Variable	SwPatchVersion	
Description	Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	**Property**	**Value**
	Type	INTEGER
	Origin	Flagchip

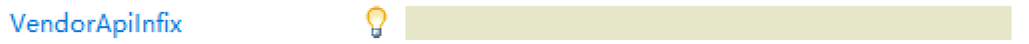
3.2.4.7 ModuleId

Variable	ModuleId	
Description	Module ID of this module from Module List.	
Screenshot		
Properties	**Property**	**Value**
	Type	INTEGER
	Origin	Flagchip

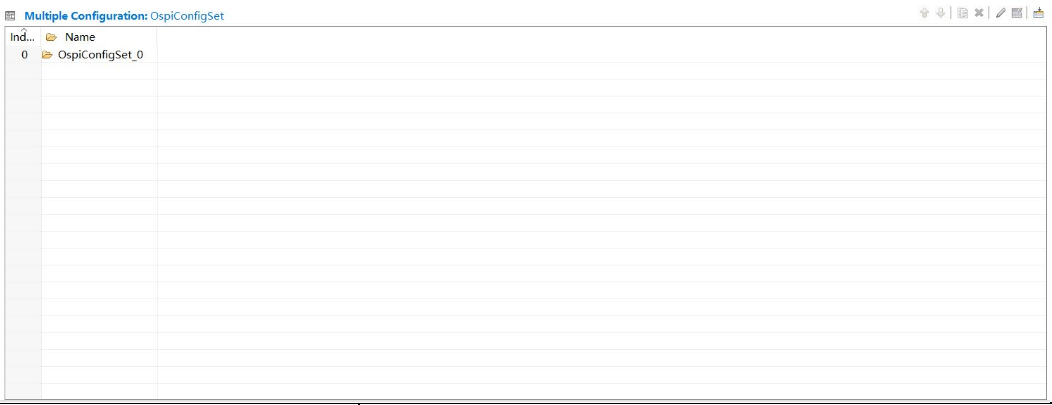
3.2.4.8 VendorId

Variable	VendorId	
Description	Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.	
Screenshot		
Properties	**Property**	**Value**
	Type	INTEGER
	Origin	Flagchip

3.2.4.9 VendorApiInfix

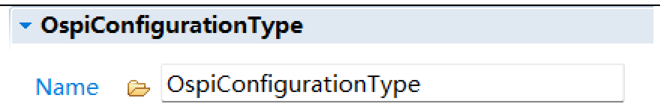
Variable	VendorApiInfix	
Description	In driver modules which can be instantiated several times on a single ECU, BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name. This parameter is used to specify the vendor specific name. In total, the Implementation specific name is generated as follows: <ModuleName>_>VendorId>_<VendorApiInfix><Api name from SWS>. E.g. assuming that the VendorId of the implementor is 123 and the implementer chose a VendorApiInfix of "v11r456" a api name Can_Write defined in the SWS will translate to Can_123_v11r456Write. This parameter is mandatory for all modules with upper multiplicity > 1. It shall not be used for modules with upper multiplicity =1. Note: Implementation Specific Parameter	
Screenshot		
Properties	**Property**	**Value**
	Type	INTEGER
	Origin	Flagchip

3.2.5 OspiConfigSet

Container	OspiConfigSet	
Description	This is the base container that contains the post-build selectable configuration parameters	
Screenshot		
Properties	**Property**	**Value**
	Type	Container

3.2.6 OspiConfigurationType

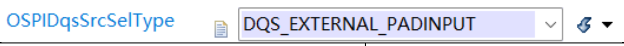
Container	OspiConfigurationType
Description	Configure ospi clock related configurations.

Screenshot		
Properties	**Property**	**Value**
	Type	Container

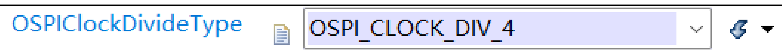
3.2.6.1 bDdrEn

Container	bDdrEn	
Description	Enables or not the DQS mode. Data source selection type for QuadSPI peripheral. Determines whether the 2x clocks are enabled for OSPI.	
Screenshot		
Properties	**Property**	**Value**
	Type	BOOLEAN
	Origin	Flagchip
	Default	FALSE

3.2.6.2 OSPIDqsSrcSelType

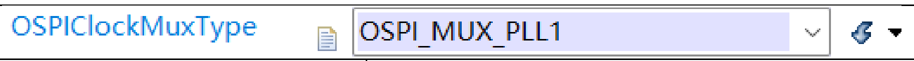
Container	OSPIDqsSrcSelType	
Description	Configure Dqs signal sources. There are three types of Dqs signal sources.	
Screenshot		
Properties	**Property**	**Value**
	Type	ENUMERATION
	Origin	Flagchip
	Default	DQS_INTER_LOOPBACK
	Range	DQS_INTER_LOOPBACK DQS_PAD_LOOPBACK DQS_EXTERNAL_PADINPUT

3.2.6.3 OSPIClockDivideType

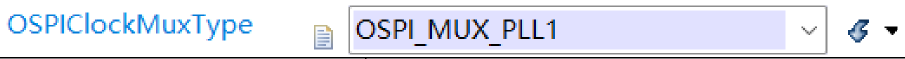
Container	OSPIClockDivideType	
Description	Prescaler factor used for generating the external clock	
Screenshot		
Properties	**Property**	**Value**
	Type	ENUMERATION
	Origin	Flagchip
	Default	OSPI_CLOCK_DIV_4
	Range	OSPI_CLOCK_DIV_1 OSPI_CLOCK_DIV_2 OSPI_CLOCK_DIV_3 OSPI_CLOCK_DIV_4 OSPI_CLOCK_DIV_5 OSPI_CLOCK_DIV_6 OSPI_CLOCK_DIV_7

		OSPI_CLOCK_DIV_8

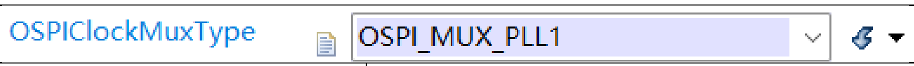
3.2.6.4 OSPIClockMuxType

Container	OSPIClockMuxType	
Description	Select the multiplex type of the clock source.	
Screenshot		
Properties	**Property**	**Value**
	Type	ENUMERATION
	Origin	Flagchip
	Default	OSPI_MUX_PLL0
	Range	OSPI_MUX_PLL0 OSPI_MUX_FIRC OSPI_MUX_PLL1 OSPI_MUX_BUSCLOCK

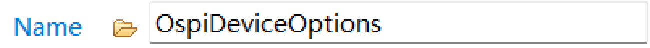
3.2.6.5 OSPIClockMuxType

Container	OSPIClockMuxType	
Description	Select the multiplex type of the clock source.	
Screenshot		
Properties	**Property**	**Value**
	Type	ENUMERATION
	Origin	Flagchip
	Default	OSPI_MUX_PLL0
	Range	OSPI_MUX_PLL0 OSPI_MUX_FIRC OSPI_MUX_PLL1 OSPI_MUX_BUSCLOCK

3.2.6.6 OSPIClockMuxType

Container	OSPIClockMuxType	
Description	Select the multiplex type of the clock source.	
Screenshot		
Properties	**Property**	**Value**
	Type	ENUMERATION
	Origin	Flagchip
	Default	OSPI_MUX_PLL0
	Range	OSPI_MUX_PLL0 OSPI_MUX_FIRC OSPI_MUX_PLL1 OSPI_MUX_BUSCLOCK

3.2.7 OspiDeviceOptions

Container	OspiDeviceOptions	
Description	Confiugration for the ospi peripherals device options	
Screenshot		
Properties	**Property**	**Value**
	Type	Container

3.2.7.1 OSPIEndianType

Container	OSPIEndianType	
Description	Select the endian type, big or small endian.	
Screenshot		
Properties	**Property**	**Value**
	Type	ENUMERATION
	Origin	Flagchip
	Default	OSPI_LITTLE_ENDIAN
	Range	OSPI_LITTLE_ENDIAN OSPI_BIG_ENDIAN

3.2.7.2 FlashAddress

Container	FlashAddress	
Description	The address of the hyperram peripheral.	
Screenshot		
Properties	**Property**	**Value**
	Type	ENUMERATION
	Origin	Flagchip
	Default	OSPI0_MEMMAP_BASE_ADDR
	Range	OSPI0_MEMMAP_BASE_ADDR

3.2.7.3 FlashTopAddress

Container	FlashTopAddress	
Description	Top address of hyperram peripherals.	
Screenshot		
Properties	**Property**	**Value**
	Type	ENUMERATION
	Origin	Flagchip
	Default	OSPI0_MEMMAP_TOP_ADDR
	Range	OSPI0_MEMMAP_TOP_ADDR

3.2.7.4 FlashColAddressSpace

Container	FlashColAddressSpace
Description	Flash column Address Space which defines the width of the column address.

Screenshot	FlashColAddressSpace (0 -> 8) 4	
Properties	**Property**	**Value**
	Type	INTEGER
	Origin	Flagchip

3.2.7.5 WordAddressable

Container	WordAddressable	
Description	Indicates whether the addressing mode is in words. If set to 0, byte addressing is indicated.	
Screenshot	WordAddressable (0 -> 1) 0	
Properties	**Property**	**Value**
	Type	INTEGER
	Origin	Flagchip

3.2.7.6 OspiTimeout

Container	OspiTimeout	
Description	Set the time for timeout. If the time is exceeded, the corresponding loop will be jumped out.	
Screenshot	OspiTimeout (0 -> 2000000) 1000000	
Properties	**Property**	**Value**
	Type	INTEGER
	Origin	Flagchip

3.2.7.7 CsHoldTime

Container	CsHoldTime	
Description	Serial flash CS hold time in terms of serial flash clock cycles.	
Screenshot	CsHoldTime (1 -> 16) 1	
Properties	**Property**	**Value**
	Type	INTEGER
	Origin	Flagchip

3.2.7.8 CsHoldTime

Container	CsHoldTime	
Description	Serial flash CS hold time in terms of serial flash clock cycles.	
Screenshot	CsHoldTime (1 -> 16) 1	
Properties	**Property**	**Value**
	Type	INTEGER
	Origin	Flagchip

3.2.7.9 CsSetupTime

Container	CsSetupTime	
Description	Serial flash CS setup time in terms of serial flash clock cycles.	
Screenshot	CsSetupTime (1 -> 16) 1	

Properties	**Property**	**Value**
	Type	INTEGER
	Origin	Flagchip

3.2.7.10 DelayLine

Container	DelayLine	
Description	Switch Delay Line into RWDS for timing. RWDS will delay for DLLINE_CFG * Tcell.	
Screenshot	DelayLine (3 -> 40) 30	
Properties	**Property**	**Value**
	Type	INTEGER
	Origin	Flagchip

3.2.8 OspiDMAConfig

Container	OspiDMAConfig	
Description	Ospi dma related configuration.	
Screenshot	Name OspiDMAConfig	
Properties	**Property**	**Value**
	Type	Container

3.2.8.1 OspiTxDmaChannel

Container	OspiTxDmaChannel	
Description	Dma channel associated to the Ospi Tx instance.	
Screenshot	OspiTxDmaChannel /DMA/DMA/DMAConfigSet/DmaHwUnit_0/DmaChannelConfig_OSPI_TX	
Properties	**Property**	**Value**
	Type	REFERENCE
	Origin	Flagchip
	REF	/DMA/DMA/DMAConfigSet/DmaHwUnit_0/DmaChannelConfig_OSPI_TX

3.2.8.2 OspiRxDmaChannel

Container	OspiRxDmaChannel	
Description	Dma channel associated to the Ospi Rx instance.	
Screenshot	OspiRxDmaChannel /DMA/DMA/DMAConfigSet/DmaHwUnit_0/DmaChannelConfig_OSPI_RX	
Properties	**Property**	**Value**
	Type	REFERENCE
	Origin	Flagchip
	REF	/DMA/DMA/DMAConfigSet/DmaHwUnit_0/DmaChannelConfig_OSPI_RX

3.2.8.3 OspiTxWatermark

Container	OspiTxWatermark
Description	Indicates the number of available spaces that will trigger a transmit action.The transmit action will be triggered if the available space (the FIFO depth minus the number of valid entries) is greater than or equal

	to the watermark value.	
Screenshot	OspiTxWatermark (0 -> 16) 8	
Properties	**Property**	**Value**
	Type	INTEGER
	Origin	Flagchip

3.2.8.4 OspiRxWatermark

Container	OspiRxWatermark	
Description	Indicates how many valid entries will trigger a readout action.The readout action will be triggered if the number of valid entries is greater than the watermark value.	
Screenshot	OspiRxWatermark (0 -> 32) 8	
Properties	**Property**	**Value**
	Type	INTEGER
	Origin	Flagchip

3.2.9 OspiNotification

Container	OspiNotification	
Description	Send notifications of status or events.	
Screenshot	Name OspiDMAConfig	
Properties	**Property**	**Value**
	Type	Container

3.2.9.1 Ospi_TransferCompleteNotification

Container	Ospi_TransferCompleteNotification	
Description	Send notifications of status or events.	
Screenshot	Ospi_TransferCompleteNotification NULL_PTR	
Properties	**Property**	**Value**
	Type	FUNCTION-NAME
	Origin	Flagchip
	Default	NULL_PTR

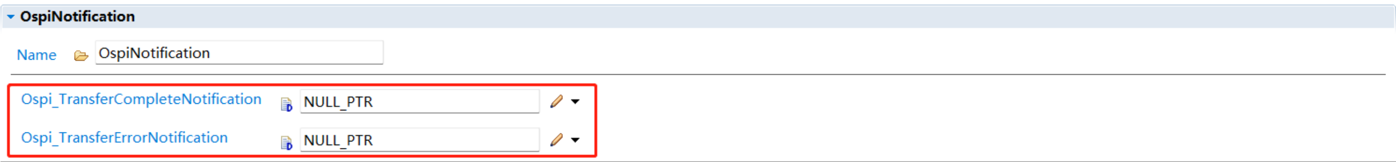
Ospi_TransferErrorNotification

Container	Ospi_TransferErrorNotification	
Description	Notify when a transmission error occurs.	
Screenshot	Ospi_TransferErrorNotification NULL_PTR	
Properties	**Property**	**Value**
	Type	FUNCTION-NAME
	Origin	Flagchip
	Default	NULL_PTR

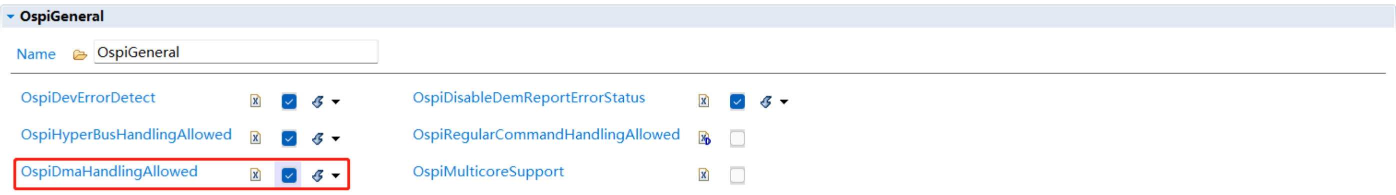
Chapter 4 Configuration Guides

4.1 Configuration Item Constraint

1) If the OSPI notification is needed, you should not set notification pointer to NULL_PTR.



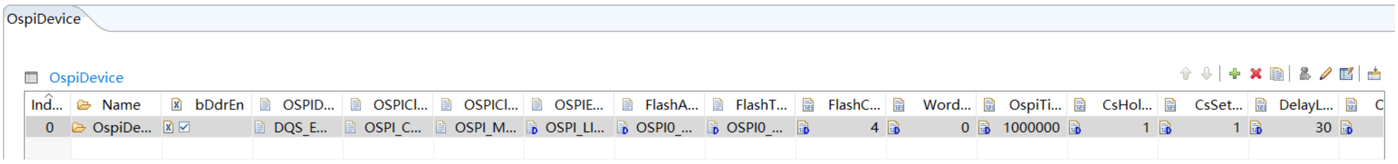
2) If DMA mode for OSPI master async transmit is needed, you should enable the item "OspiDmaHandlingAllowed".



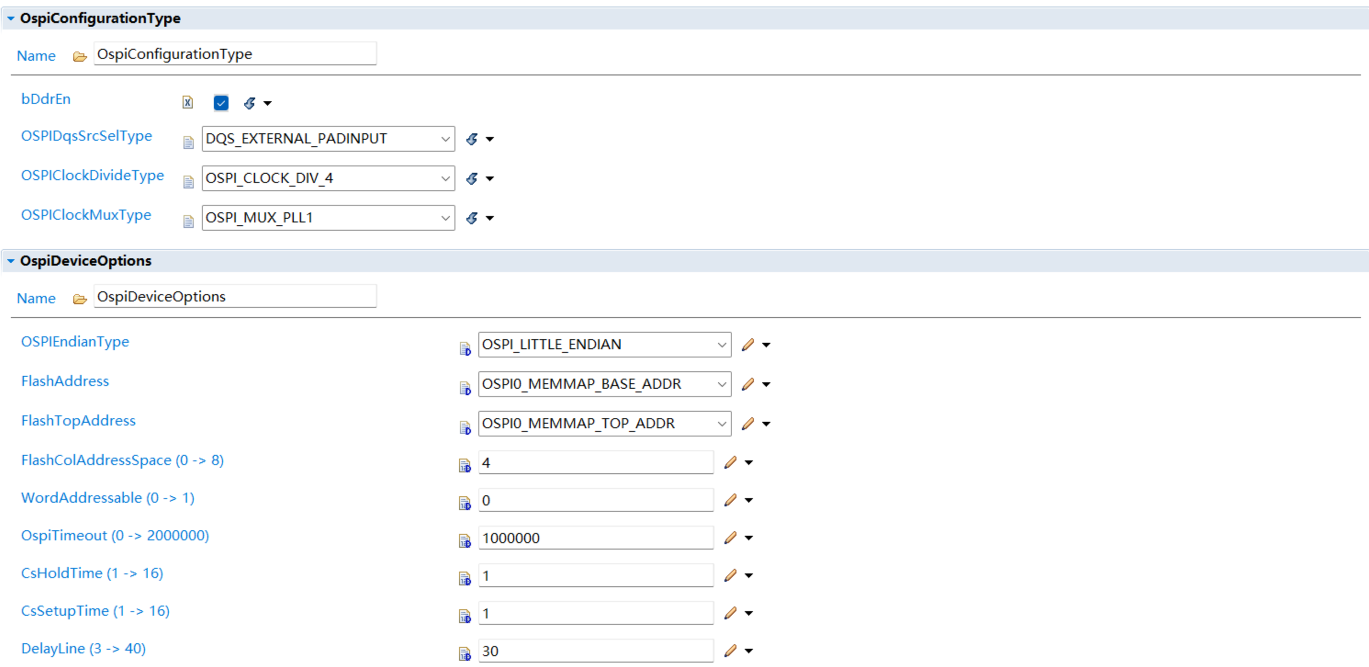
4.2 OSPI Usage Common Steps

Basically, the OSPI module could be configured by following the below steps:

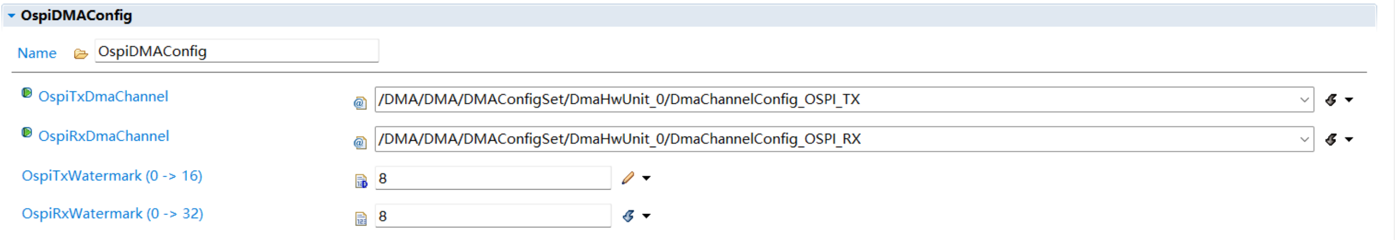
1) Configure OSPI device in the tab "OspiDevice".



2) Double click item, select OSPI related parameters.



3) If master mode and item "OspiDmaHandlingAllowed" is enable, you can configure the DMA channel and watermark value.



4) Generate configuration files.